

Santiment Social Volume Intraday Seasonality

A Baseline for Crypto Attention

Metric: Santiment social volume

Sample: 2025-06-20 through 2026-05-19, complete UTC days

Assets: bitcoin, ethereum, solana, xrp, dogecoin, binance-coin, cardano, chainlink, avalanche, tron

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Executive Summary

The result is not subtle. Social volume in large-cap crypto assets has a pronounced intraday rhythm: the strongest UTC hour is 14:00, the weakest is 04:00, and the average peak-to-trough spread is 2.21x. A raw social-volume print is therefore not a clean abnormality signal unless the clock is part of the model.

The weekly effect is smaller but still operationally material. Weekend daily social volume is 25.3% lower than weekdays in this sample. This is large enough to distort alert thresholds, cross-asset dashboards, and manual anomaly review.

The practical rule is simple: treat hour-of-day and weekday seasonality as the first risk control in any short-horizon social-volume workflow. This is an attention-baseline study, not a claim about return predictability.

The Question

Crypto trades continuously. Human attention does not. The question is whether the timing of social attention is stable enough, and large enough, to deserve explicit treatment in monitoring infrastructure.

We test a narrow question: **how much of observed social-volume variation is explained by recurring UTC hour and weekday effects, before making any claim that activity is abnormal?**

Relation to Prior Evidence

This note sits at the intersection of two established research lines. The first is the investor-attention literature: Da, Engelberg, and Gao (2011) formalize internet search activity as a direct attention proxy and show that attention can be distinct from standard market observables. The implication for social volume is clear: attention measures can be economically meaningful precisely because they capture a different state variable, but that state variable still needs its own baseline.

The second line is high-frequency market seasonality. Andersen and Bollerslev (1997) show that intraday periodicity can dominate high-frequency dynamics unless explicitly modeled; ignoring the calendar can make ordinary periodic structure look like persistence or abnormality. Crypto evidence points in the same direction: Eross et al. (2019) document systematic intraday variation in Bitcoin volume, volatility, liquidity, and returns. Our contribution is to apply this baseline logic to crypto social attention rather than prices or trading volume.

Crypto-specific social-data studies provide the motivation, but also the guardrail. Garcia et al. (2014) link word-of-mouth communication, information search, and Bitcoin market dynamics in feedback cycles. Shen, Urquhart, and Wang (2019) find that Twitter activity is a significant driver of next-day trading volume and realized volatility. These papers justify taking social activity seriously, especially as an activity and risk-state variable. They do not remove the need to separate predictable calendar structure from genuine information shocks, and they do not turn raw social volume into a directional return signal.

Data and Estimation

We study hourly Santiment social volume for a liquid large-cap panel: bitcoin, ethereum, solana, xrp, dogecoin, binance-coin, cardano, chainlink, avalanche, and tron. The sample contains complete UTC days from 2025-06-20 through 2026-05-19.

For each asset-day, hourly social volume is converted into a daily-share index:

$$I_{a,d,h} = 24 \times \frac{V_{a,d,h}}{\sum_{j=0}^{23} V_{a,d,j}}$$

An index of 1.10 means that an hour receives 10% more social volume than an equal 1/24 daily allocation. This normalization is deliberately conservative: it preserves the intraday profile while reducing contamination from asset scale, secular attention growth, and idiosyncratic news days.

Uncertainty bands use a date-block bootstrap. Resampling dates, rather than asset-hour observations, preserves same-day cross-asset dependence.

Sample Coverage

Coverage item	Value
Assets	10
Complete asset-days	3,340
Hourly observations after complete-day filter	80,160
Expected hours per asset	8,016

Top assets by sample social volume:

Asset	Hours	Coverage (%)	Total (k)	Mean per hour
bitcoin	8016	100.00	1394.9	174.02
ethereum	8016	100.00	427.0	53.27
solana	8016	100.00	263.9	32.93
xrp	8016	100.00	120.4	15.03
binance-coin	8016	100.00	63.4	7.91
dogecoin	8016	100.00	57.9	7.22
cardano	8016	100.00	43.6	5.44
tron	8016	100.00	38.7	4.83
chainlink	8016	100.00	25.6	3.20
avalanche	8016	100.00	21.4	2.67

The panel is intentionally liquid and large-cap. It is designed to estimate the timing structure of major-asset attention, not the average behavior of the full crypto asset universe.

Result 1: The Clock Dominates the Baseline

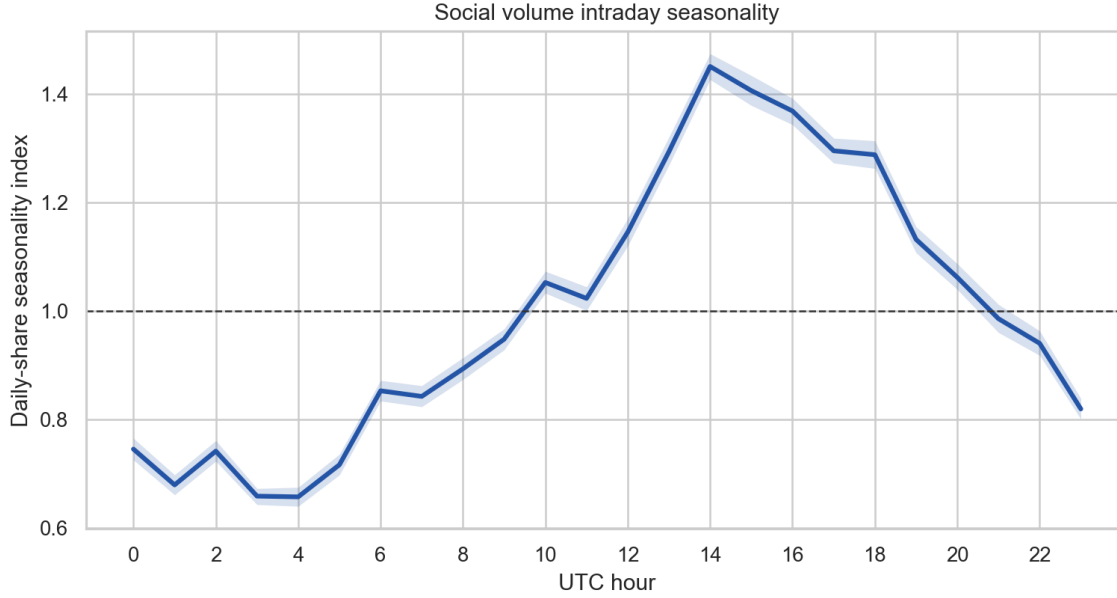


Figure 1: Hourly seasonality index

Top UTC hours by daily-share index:

UTC hour	Mean index	95% confidence interval
14	1.451	[1.427, 1.474]
15	1.406	[1.379, 1.434]
16	1.369	[1.344, 1.392]
17	1.295	[1.273, 1.319]
13	1.294	[1.269, 1.321]

Lowest UTC hours by daily-share index:

UTC hour	Mean index	95% confidence interval
4	0.657	[0.640, 0.675]
3	0.659	[0.643, 0.673]
1	0.679	[0.661, 0.698]
5	0.716	[0.698, 0.736]
2	0.742	[0.723, 0.761]

The interpretation is mechanical but important. A social-volume observation during the global attention window should not be benchmarked against the same raw value during the quiet UTC hours. A fixed threshold is a clock bet disguised as an anomaly detector.

Result 2: The Pattern Is Not a Bitcoin Artifact

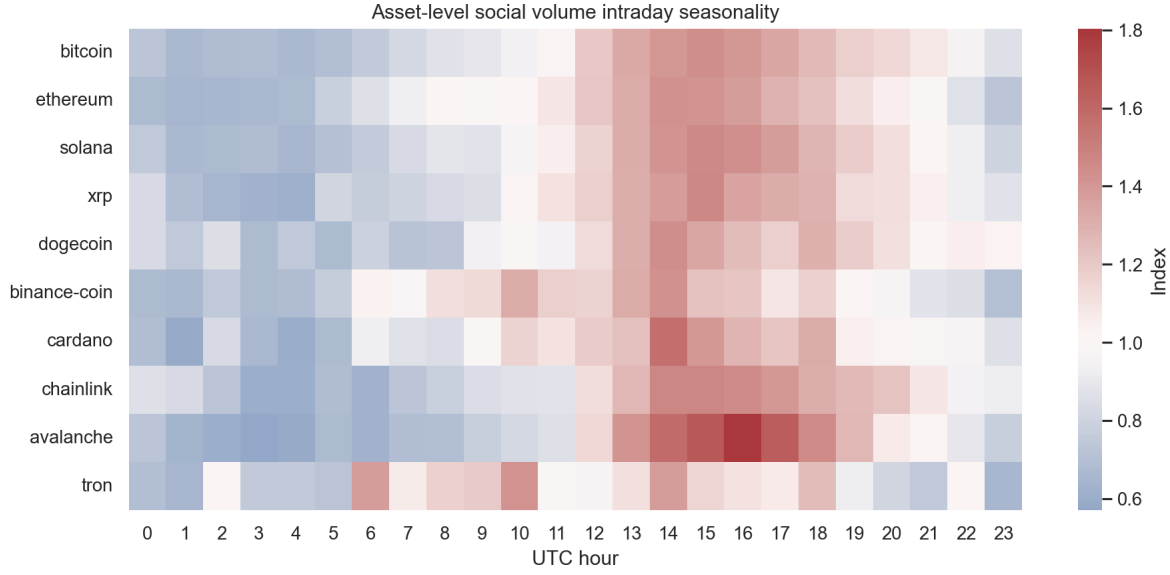


Figure 2: Asset-level intraday index

The asset-level heatmap shows a common intraday profile across the panel. Event-driven dispersion is higher for larger attention assets, but the core timing pattern is broad. This makes the adjustment suitable as a panel-level baseline rather than a one-off BTC correction.

Result 3: Weekends Carry a Lower Prior

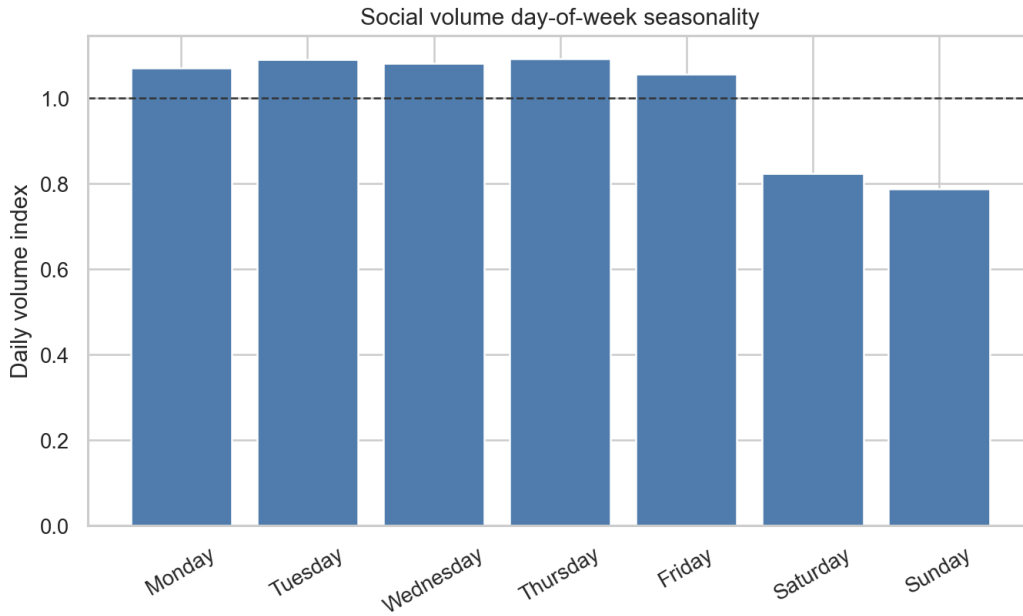


Figure 3: Day-of-week index

Day	Mean index	Median index	Asset-days
Monday	1.071	0.967	480
Tuesday	1.091	0.972	480
Wednesday	1.082	0.978	470
Thursday	1.092	0.952	470
Friday	1.057	0.927	480

Day	Mean index	Median index	Asset-days
Saturday	0.823	0.724	480
Sunday	0.788	0.687	480

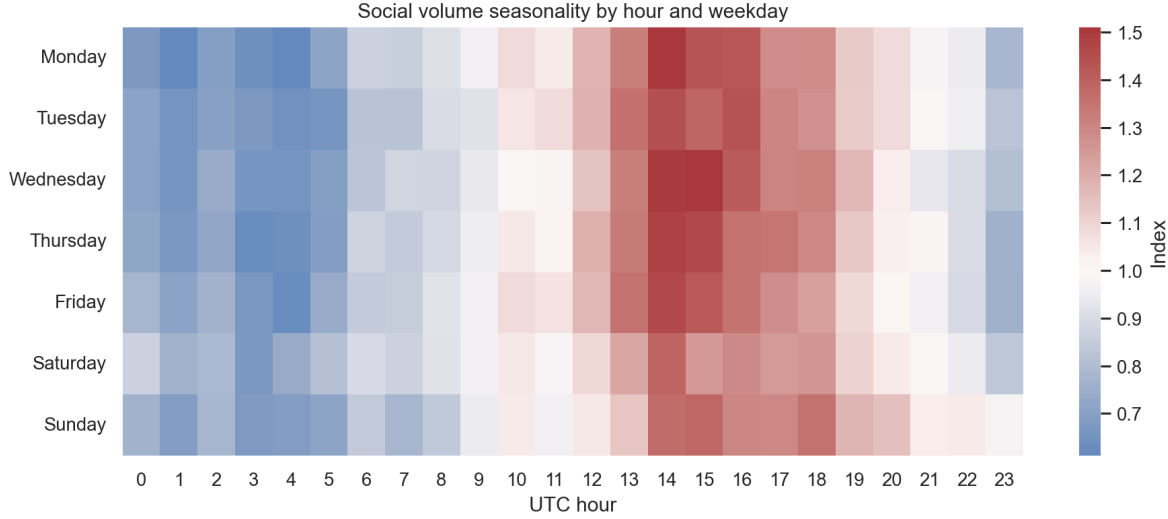


Figure 4: Hour by day-of-week heatmap

The day-of-week effect is less dramatic than the hour-of-day effect, but it is too large to ignore. A Saturday print and a Tuesday print at the same UTC hour do not have the same prior.

Case Study: A Seasonality-Adjusted Alert

Consider a BTC social-volume print at 14:00 UTC. A raw threshold may classify the observation as high. But 14:00 UTC is also the strongest hour in the baseline. The operational question is not whether the print is large in absolute terms; it is whether the print is large relative to what should have happened for that asset, hour, and weekday.

The practical solution is to smooth the expected baseline and the residual, not the raw social-volume series. Smoothing raw activity directly risks suppressing the information shock we are trying to detect. The clock should be removed before the signal is judged.

A production implementation can be kept deliberately simple:

1. **Estimate a smoothed clock baseline.** For each asset, weekday, and UTC hour, estimate a historical daily-share index. To reduce noise, shrink the asset-specific clock toward the panel clock:

$$\hat{c}_{a,w,h} = \lambda_a c_{a,w,h}^{asset} + (1 - \lambda_a) c_{w,h}^{panel}$$

where λ_a increases with the asset's history and social-volume depth. Then apply light circular smoothing across adjacent hours and renormalize the 24 hourly weights so that their average is one.

2. **Estimate the current daily attention level.** Use an exponentially weighted moving average of recent complete days for the same asset. This captures whether the asset is in a high-attention or low-attention regime before allocating that daily expectation across the clock.
3. **Convert the baseline into an expected hourly value.**

$$\hat{V}_{a,t} = \hat{D}_{a,t} \times \frac{\hat{c}_{a,w(t),h(t)}}{24}$$

where $\hat{D}_{a,t}$ is the estimated daily attention level and $\hat{c}_{a,w(t),h(t)}$ is the smoothed clock weight for the current weekday and hour.

4. **Score the observation as a residual.** A log residual is more stable than a raw difference:

$$r_{a,t} = \log(1 + V_{a,t}) - \log(1 + \widehat{V}_{a,t})$$

Standardize this residual with a robust estimate of historical residual volatility for the same asset and clock bucket:

$$z_{a,t} = \frac{r_{a,t}}{\widehat{\sigma}_{a,w(t),h(t)}}$$

5. **Smooth the residual signal.** Apply a short half-life filter to the standardized residual:

$$\tilde{z}_{a,t} = \alpha z_{a,t} + (1 - \alpha) \tilde{z}_{a,t-1}$$

A half-life of 2-3 hours is usually enough to reduce one-hour noise without erasing a persistent information event.

The resulting alert should expose four fields: observed social volume, expected social volume for the asset-hour-weekday bucket, the seasonality-adjusted ratio, and the smoothed residual z-score. In this framing, the alert is not “social volume is high.” It is “social volume is high after removing the clock.”

Implications

1. **Seasonality adjustment is not cosmetic.** It is the difference between measuring abnormal attention and measuring the clock.
2. **Raw cross-asset comparisons are scale comparisons.** BTC and ETH dominate the level of social volume. For monitoring, the relevant object is abnormality relative to the asset’s own baseline.
3. **Alerts should expose the baseline, not only the print.** A useful alert should report the observed value, the expected value for that hour and weekday, and the residual.
4. **This is infrastructure for research, not a standalone alpha claim.** Prior crypto-attention evidence is strongest around activity, feedback, volume, and volatility. Seasonality-adjusted social volume may be an input into signal research, but this note tests baseline structure only.

Limits of the Evidence

- The panel covers ten liquid large-cap assets, not the full Santiment asset universe.
- The analysis is in UTC. Desk-specific deployment may require local-session overlays.
- The measure captures message volume, not sentiment, author uniqueness, engagement quality, or source composition.
- The sample is historical and should be refreshed before time-sensitive publication or client-specific implementation.
- No return forecast is estimated in this note.

Conclusion

Social volume is seasonal enough that the burden of proof should shift. The default assumption should not be that a raw print is abnormal; it should be that the print is first explained by hour, weekday, and asset baseline. Only the residual deserves escalation.

References

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